

BONUS 1 OF 4

+1

Luau Cheat Sheet

YOU GET

The syntax, services and patterns every Roblox game uses.

Everything you'll type 90% of the time, on a few pages. Keep it open next to Roblox Studio.

Basics

```
local name = "Coin Rush"      -- string
local coins = 0                -- number
local alive = true             -- boolean
local nothing = nil            -- no value

coins += 10                    -- also -=, *=, /=
local label = `Coins: {coins}` -- string interpolation
print(#name)                   -- length of a string or list: 9
```

Compare	Meaning	Logic	Meaning
<code>==</code>	equal	<code>and</code>	both true
<code>~=</code>	not equal	<code>or</code>	either true
<code><</code> <code>></code> <code><=</code> <code>>=</code>	order	<code>not</code>	flip

Control flow

```
if coins >= 100 then
    print("Rich")
elseif coins > 0 then
    print("Getting there")
else
    print("Broke")
end

for i = 1, 10 do print(i) end      -- 1..10
for i = 10, 1, -1 do print(i) end -- countdown
for index, value in list do end   -- lists
for key, value in dict do end     -- dictionaries

while running do
    task.wait(1)                  -- ALWAYS wait inside a loop
end
```

Functions

```
local function add(a: number, b: number): number
    return a + b
end

local ok, result = pcall(riskyFunction)  -- catch errors
task.spawn(fn)                          -- run without waiting
task.delay(2, fn)                       -- run after 2 seconds
```

Tables

```
local list = { "Red", "Blue" }
table.insert(list, "Green")
table.remove(list, 1)
print(list[1], #list)

local prices = { Speed = 100, Jump = 150 }
prices.Shield = 500
prices.Speed = nil  -- delete a key
local copy = table.clone(prices)
```

Services you'll use constantly

```
local Players = game:GetService("Players")
local ReplicatedStorage = game:GetService("ReplicatedStorage")
local ServerStorage = game:GetService("ServerStorage")
local TweenService = game:GetService("TweenService")
local RunService = game:GetService("RunService")
local DataStoreService = game:GetService("DataStoreService")
local MarketplaceService = game:GetService("MarketplaceService")
local SoundService = game:GetService("SoundService")
local Debris = game:GetService("Debris")
```

Instances

```
local part = Instance.new("Part")
part.Size = Vector3.new(4, 1, 4)
part.Position = Vector3.new(0, 10, 0)
part.Anchored = true
part.Color = Color3.fromRGB(198, 255, 61)
part.Material = Enum.Material.Neon
part.Parent = workspace          -- set Parent LAST

local obj = workspace:WaitForChild("Lava")    -- waits until it exists
local maybe = workspace:FindFirstChild("Lava") -- nil if missing
obj:Destroy()
```

Players & characters

```
Players.PlayerAdded:Connect(function(player)
    player.CharacterAdded:Connect(function(character)
        local humanoid = character:WaitForChild("Humanoid")
        humanoid.WalkSpeed = 20
    end)
end)

local player = Players:GetPlayerFromCharacter(hit.Parent) -- who touched?
local me = Players.LocalPlayer                          -- client only
```

Patterns you'll reuse forever

Debounce (stop an event firing 30× a second)

```
local busy = false
part.Touched:Connect(function(hit)
    if busy then return end
    busy = true
    -- do the thing
    task.wait(1)
    busy = false
end)
```

Secure remote (client asks, server decides)

```
Remote.OnServerEvent:Connect(function(player, itemName)
    if typeof(itemName) ~= "string" then return end
    local price = Prices[itemName]
    if not price or coins.Value < price then return end
    coins.Value -= price
end)
```

Tween (smooth animation)

```
TweenService:Create(frame, TweenInfo.new(0.3, Enum.EasingStyle.Back),
    { Position = UDim2.fromScale(0.5, 0.5) }):Play()
```

Save data safely

```
local ok, err = pcall(function()
    store:UpdateAsync("Player_" .. player.UserId, function()
        return data
    end)
end)
if not ok then warn("Save failed:", err) end
```

UI quick reference

Want	Use
Same size on every screen	<code>UDim2.fromScale(0.2, 0.1)</code>
Center something	<code>AnchorPoint = Vector2.new(0.5, 0.5)</code> , <code>Position = UDim2.fromScale(0.5, 0.5)</code>
Keep a shape square	<code>UIAspectRatioConstraint</code> with <code>AspectRatio = 1</code>
Rounded corners	<code>UICorner</code>
Outline text	<code>UIStroke</code>
Stack items	<code>UIListLayout</code>
Button that works on touch + mouse + gamepad	<code>button.Activated:Connect(...)</code>

Where scripts go

Script type	Put it in	Runs on
Script	ServerScriptService	Server
LocalScript	StarterPlayerScripts, StarterGui, ReplicatedFirst	Player's device
ModuleScript	ReplicatedStorage (shared) or ServerScriptService (server-only)	Whoever requires it

TIP When something doesn't work, open **View** → **Output** first. The error message and line number tell you where to look 90% of the time.